A 3×3 Ordered Binary Matrix Removal Obstruction

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Abstract

Candidate result with a Lean-checked qualitative theorem. The forbidden pattern has nine entries, down from twelve in our previous 3×4 construction. We retain arbitrary binary repairs, arbitrarily small distance parameters, and arbitrarily large hosts at the same chosen parameter. Independent mathematical peer review is pending; no global minimality or worldwide priority is claimed.

The written construction below has $n = (8h + 2)m$ and exactly $7m^4$ copies. A separate AI reviewer found no gap in this argument. The Lean proof establishes the same qualitative theorem using doubled anchor blocks, $n = (14h + 2)m$, and the sufficient bound $2mn^4$. It does not certify the written exact count or smaller host size. Section 8 specifies the formal statement and evidence.

Background and relation to prior work

Alon, Fischer and Newman proved polynomial removal bounds for binary matrices when copies are considered up to row and column permutations [1]. Alon, Ben-Eliezer and Fischer established a qualitative removal lemma for ordered matrices and constant-query testing of hereditary matrix properties [2]. The polynomial bound of Alon and Ben-Eliezer for hitting all existing copies [3, Theorem 1.2] addresses a different quantity from binary repair distance: editing a hitting set can create new copies. Here copies preserve the separate row and column orders, and repairs must eliminate all copies while permitting changes in either binary direction.

OpenAI gave a fixed 66×66 obstruction to polynomial ordered binary matrix removal [4]. Subsequent repository manuscripts include the ProveIt project's 5×5 construction [5], Erebos Labs' 4×4 construction [6], and Wang's 4×4 construction with exact copy and repair counts [7]. The present 3×3 construction continues our preceding 3×4 refinement. The bibliography identifies the versions used; BASELINE.md records their distinct proof and formal-verification scopes.

1 Statement and quantitative implication

Let

$$K = J_3 - I_3 = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}.$$

For every integer $h \geq 1$, put $m = 2^h$, $d = 8h + 2$ and $n = dm$. The square binary host A_h defined below satisfies

$$D_K(A_h) \geq m^2, \qquad N_K(A_h) = 7m^4, \qquad \frac{N_K(A_h)}{n^6} = 7d^{-6}4^{-h}.$$

D_K minimizes **all binary edits in both directions, at arbitrary cells**. Copies preserve independent increasing row and column orders and match all nine entries. No exact minimum repair distance is claimed.

Consequently, for every $c, C, \epsilon_0 > 0$ there exists a fixed $0 < \epsilon < \epsilon_0$ such that, for every N, some $n \geq \max(1, N)$ and binary $n \times n$ host require at least ϵn^2 edits to become K-free and have fewer than $c\epsilon^C n^6$ copies. This is a 3×3 pattern, with 9 rather than the preceding 3×4 pattern's 12 entries.

Bitwise complementation gives the same statement for the **3×3 identity matrix**: every K-copy in A corresponds to an I_3-copy in $1 - A$, and $B \mapsto 1 - B$ preserves edit distance and exchanges the two freeness conditions.

2 Complete host specification

At each depth $0 \leq i < h$ and node $p \in \{0, \ldots, 2^i - 1\}$, create blocks $R_i^-(p)$, $R_i^+(p)$, $C_i^-(p)$, $C_i^+(p)$, each of size $m/2^i$. At depth h, identify both signs with leaf blocks $L_R(p)$, $L_C(p)$, of size one. Create dummy blocks R_*, C_* of size m.

The recursive row order is minus block, left subtree, right subtree, plus block. The column order is plus block, left subtree, right subtree, minus block. Dummy rows follow all variable rows; dummy columns precede all variable columns. Every nonanchor entry incident to a dummy is one. All variable entries are zero except these threshold entries:

Row / column	Entry	Range
$R_i^+(p), C_i^+(q)$	$1[p \leq q]$	$0 \leq i \leq h$
$R_i^-(p), C_i^-(q)$	$1[p < q]$	$0 \leq i < h$
$R_i^+(p), C_{i-1}^+(q)$	$1[\lfloor p/2 \rfloor \leq q]$	$1 \leq i \leq h$
$R_i^-(p), C_{i-1}^-(q)$	$1[\lfloor p/2 \rfloor < q]$	$1 \leq i \leq h$

At $i = h$ the first rule means the leaf-to-leaf matrix, and both parent rules apply to the shared leaf row. Write $\pi(p) = \lfloor p/2 \rfloor$. A class is the union over all nodes at a fixed signed depth; a parity subscript restricts the node parity.

There are six modes at each $i = 1, \ldots, h$:

Mode	Row role 1	Row role 2	Column role 1	Column role 2
V_i^+	R_i^+	R_{i-1}^+	C_*	C_{i-1}^+
V_i^-	R_{i-1}^-	R_i^-	C_*	C_{i-1}^-
$W_{i,e}^+$	$R_{i,e}^+$	R_*	C_{i-1}^+	$C_{i,e}^+$
$W_{i,e}^-$	$R_{i,e}^-$	R_*	$C_{i,e}^-$	C_{i-1}^-

Add one row block X_t and one column block Y_t of size m per mode t. All X blocks precede all nonanchor rows. All Y blocks follow all nonanchor columns. The mode order on each axis is W^- in increasing (i, e), then W^+ in increasing (i, e), then $V_1^+, V_1^-, \ldots, V_h^+, V_h^-$.

Set $A(X_s, Y_t) = 1[s = t]$. On nonanchor rows, the one-set of Y_t is exactly

$$U_t = (\text{row role 1 of } t) \ \cup \ \begin{cases} R_* & t \text{ vertical,} \\ \varnothing & t \text{ horizontal.} \end{cases}$$

The one-set T_t of X_t on nonanchor columns is:

t	T_t
V_i^σ	C_{i-1}^σ
$W_{i,e}^+$	$C_* \cup C_{i,e}^+$
$W_{i,1}^-$	$C_* \cup C_{i-1}^-$
$W_{i,0}^-$, $i < h$	$C_* \cup C_{i-1}^- \cup C_{i,1}^-$
$W_{h,0}^-$	$C_* \cup C_{h-1}^-$

All unspecified entries are zero. In particular, the odd-child addition is absent at the shared leaves. Each axis has $6hm$ anchor positions and $(2h+2)m$ nonanchor positions, so $n = (8h+2)m$.

On the two column roles X_t has signature 01; on the two row roles Y_t has signature 10. The added dummy one for vertical Y_t is outside both row roles. Thus every ordered role body $P = ((1,0),(1,1))$ extends to K using X_t as the first row and Y_t as the last column.

3 Elementary order and support facts

Blocks in a fixed class occur in increasing node order. A fixed nonanchor row is nondecreasing along a column class; a fixed nonanchor column is nonincreasing along a row class. These follow from the four thresholds. For the comparisons used below, the recursive order gives

$$\begin{aligned} C_{i-1}^+(a) < C_i^\sigma(q) &\iff a \le \pi(q), \\ C_i^\sigma(q) < C_{i-1}^-(b) &\iff \pi(q) \le b, \\ C_{i-1}^-(b) < C_i^\sigma(q) &\iff b < \pi(q), \\ C_j^+(a) < C_j^-(b) &\iff a \le b, \\ R_i^+(p) < R_{i-1}^+(a) &\iff \pi(p) \le a, \\ R_{i-1}^-(a) < R_i^-(p) &\iff a \le \pi(p). \end{aligned}$$

Child signs can be the common leaf sign; the same-level cross-sign comparison has $j < h$. Reverse comparisons have the complementary inequalities.

Nonanchor crown exclusion. A dummy cannot occur in a crown, since each crown row and column contains a zero. Two crown columns cannot be in one variable class, because their one-sets would be nested. The classes on which a nondummy row can be one are its own and preceding signed classes, or, for a leaf row, $L_C, C_{h-1}^+, C_{h-1}^-$. Three distinct column classes with a common nondummy neighbor for each pair must therefore be this last triple: elsewhere the class-incidence graph is two paths. The two crown rows incident to the leaf column must both be leaves. Their nonanchor neighborhoods are nested as the leaf index increases, contradicting the crown's incomparable row traces. This excludes every row/column permutation of the crown in the nonanchor body.

Crossing anchor supports. Suppose modes $A < B$ have ordered variable columns $u < v < w$ with

$$u \in T_B \setminus T_A, \quad v \in T_A \setminus T_B, \quad w \in T_A \cap T_B.$$

Then $A = W^-_{i,0}$ for $i < h$, and B is $W^-_{i+1,e}$ or V^-_{i+1}. Moreover $v \in C^-_{i-1}$, $w \in C^-_{i,1}$, and u is in $C^-_{i,0}$ or, when $B = W^-_{i+1,0}$ and $i + 1 < h$, in $C^-_{i+1,1}$.

Here is an exhaustive justification from the support table. At one minus level the supports are nested. Distinct minus levels have a variable intersection only at successive levels with the earlier mode's odd-child addition. A minus mode can cross a later vertical support only when that support is its full child class. Plus horizontal supports at distinct levels have no variable intersection; at one level they are disjoint apart from C_*. A plus horizontal support and a later vertical support can cross only in the former's parity class, leaving C_* as the first support's exclusive part. It cannot be v, since $u < v$ and C_* is first. Vertical supports are distinct full classes. Plus and minus variable supports do not intersect: the minus odd-child addition never reaches the shared leaf class. These exhaust the mode types and their order.

4 Every copy has exactly one anchor on each axis

Write a copy on $r_1 < r_2 < r_3$ and $c_1 < c_2 < c_3$. Let a, b count its anchor rows and columns. Rows and columns of K are pairwise distinct, so no twin block can be used twice. Row anchors form a prefix of the copy; column anchors form a suffix.

$a = 0$. If $b = 0$, use nonanchor crown exclusion. Otherwise $c_3 = Y_t$. Its ones on r_1, r_2 put them in the same variable class: neither can be the last nonanchor row type, the dummy. If $b = 1$ or 2, c_1 is nonanchor, and its trace 01 on $r_1 < r_2$ contradicts nonincreasing entries within their row class. If $b = 3$, c_1 and c_2 have opposite traces 01 and 10 on those two rows. On one variable class a U support is empty, full, or one parity half. Thus these two supports must be the two opposite parity halves, both belonging to horizontal modes. They are disjoint even on the dummy, contradicting their two ones on r_3.

$a \geq 1, b \geq 2$. Row r_1 is an X block and is one on both c_2, c_3. It is one on only one Y block, so these columns would be twins.

$a \geq 2, b = 1$. Column c_3 is one on both r_1, r_2. Its unique matching X block makes these rows twins.

$a \geq 2, b = 0$. Put $(u, v, w) = (c_1, c_2, c_3)$. The common w and the middle v cannot be dummy because earlier columns exist. If u were dummy, the earlier anchor would be vertical and the later one horizontal, contrary to the global mode order. Thus all three columns are variable, and the first two supports give the crossing configuration of Section 3. If $a = 2$, r_3 is nondummy (it is zero on w). It must be one on u, v. When u is at level $i + 1$, the two levels $i + 1$ and $i - 1$ are too far apart for one nonanchor row support. When u is in even C^-_i, r_3 must be in R^-_i. Since $u < w$ within that class, its one on u forces a one on w, a contradiction. If $a = 3$, no anchor support contains both C^-_{i-1} and C^-_{i+1}. The only anchor support spanning C^-_{i-1} and C^-_i adds odd children, not the required even u. This is also impossible.

We are left with $a = 1$ and $b \leq 1$. Suppose $b = 0$. Now c_2, c_3 lie in T_t for $r_1 = X_t$. They are variable, since a smaller c_1 exists. If they belong to the same class, the trace 10 of r_3 on $c_2 < c_3$ is impossible. The only remaining support is $T_{W^-_{i,0}}$, $i < h$, and their classes are C^-_{i-1}, C^-_i in some order. The common positive column c_1 of r_2, r_3 is not dummy, since this T_t contains C_*. Both lower rows are nondummy. Inspection of their own/parent supports puts the class of c_1 in $\{C^-_{i-1}, C^-_i\}$: a row incident to the former has depth $i - 1$ or i, and a row incident to the latter has depth i or $i + 1$;

intersections have no third class, including when $i+1=h$. If c_1 has c_2's class, row r_2 has forbidden trace 10 on $c_1 < c_2$. If it has c_3's class, row r_3 has that trace on $c_1 < c_3$. Therefore $b = 1$.

5 Localization and exact enumeration

We now have $r_1 = X_t$ and $c_3 = Y_t$, with the same mode because their corner entry is one. Row r_2 is variable, lies in row role 1 of t, and satisfies

$$c_1 \notin T_t, \quad c_2 \in T_t, \quad A(r_2, c_1) = 1, \quad A(r_2, c_2) = 0.$$

The later row r_3 is one on both nonanchor columns and zero on Y_t.

$\boldsymbol{t = V_i^+}$**.** Here $r_2 = R_i^+(p)$, $c_2 = C_{i-1}^+(q)$, and $\pi(p) > q$. Since r_3 is zero on Y_t, it is neither dummy nor in R_i^+. Its one on c_2 forces $r_3 = R_{i-1}^+(a)$. Row order gives $\pi(p) \le a$, while that one gives $a \le q$, a contradiction.

$\boldsymbol{t = V_i^-}$**.** Now $r_2 = R_{i-1}^-(p)$, $c_2 = C_{i-1}^-(q)$, $p \ge q$. The zero on Y_t excludes the dummy and R_{i-1}^-, so r_3 must be $R_i^-(a)$, allowing a leaf. Row order and its one on c_2 give $p \le \pi(a) < q$, a contradiction. These two exclusions use the new dummy ones on vertical Y blocks.

$\boldsymbol{t = W_{i,e}^+}$**.** The row r_2 and column c_2 are in the same plus child class and parity; their nodes p, q satisfy $p > q$. A positive c_1 outside T_t cannot be dummy. If c_1 is in the child class, monotonicity contradicts $c_1 < c_2$. If it is a plus parent with node u, positivity and order give $\pi(p) \le u \le \pi(q)$, impossible for $p > q$ of equal parity. At the leaf level a minus parent is additionally possible, but then $\pi(p) < u < \pi(q)$, also impossible.

$\boldsymbol{t = W_{i,e}^-}$**.** Write $r_2 = R_i^-(p)$, or $L_R(p)$ when $i = h$. If c_2 is an added odd child, then $i < h$, $e = 0$, and its node q satisfies $p \ge q$. The only positive c_1 outside T_t is an own-level node $u > p$; its order before c_2 gives $u < q$, impossible. Thus $c_2 = C_{i-1}^-(b)$ and $\pi(p) \ge b$.

For $i < h$, a positive c_1 outside T_t must be $C_i^-(q)$, with $q > p$. Its position before c_2 gives $\pi(q) \le b$. Thus $\pi(q) = b = \pi(p)$, p is even, and $q = p+1$ is odd. But this odd child belongs to T_t, a contradiction. This is the precise use of the nonterminal odd-child addition.

Consequently $i = h$. Column c_1 is either a leaf $q \ge p$ or a plus-parent $C_{h-1}^+(u)$ with $\pi(p) \le u$. Its order before c_2 gives $\pi(q) \le b$ in the first case and $u \le b$ in the second. Combining with $\pi(p) \ge b$ forces

$$b = \pi(p), \qquad c_1 \in \{L_C(q) : p \le q,\ \pi(q) = \pi(p)\} \ \cup\ C_{h-1}^+(\pi(p)).$$

Finally r_3 must be dummy. A variable row one on $C_{h-1}^-(\pi(p))$ is either a leaf v or $R_{h-1}^-(a)$. A later leaf has $v > p$ and cannot satisfy $\pi(v) < \pi(p)$. A later parent-minus row has $a > \pi(p)$ and cannot satisfy $a < \pi(p)$. No other variable row can be one there.

We have classified every copy: for a leaf p, it uses $X_{W_{h,p \bmod 2}^-}$, $L_R(p)$, R_*; $C_{h-1}^-(\pi(p))$; the matching Y block; and either a leaf $q \ge p$ in that parent's two-leaf interval or $C_{h-1}^+(\pi(p))$ as its first column. Conversely these choices have the required order and all nine entries, directly by the host rules.

Three anchor/dummy choices contribute m^3, and the minus-parent column contributes 2. The first column has 2 plus-parent choices and 2 leaf choices for even p, or 2 plus-parent choices and 1 leaf choice for odd p. There are $m/2$ leaves of each parity, so

$$N_K = \frac{m}{2} \cdot 2m^3(4+3) = 7m^4.$$

The two families contribute $4m^4$ and $3m^4$ respectively. In particular every copy's middle zero lies in the $2m$-cell leaf-to-own-minus-parent set.

6 Lower bound against arbitrary repairs

Let a binary B differ from A_h on $M < m^2$ cells. Sample a uniform leaf z and its ancestor path. Independently choose a uniform representative from each signed row and column block on the path, identifying the two sign choices at the shared leaf on each axis. Choose independent uniform representatives from every anchor and dummy block.

Every variable or dummy position has marginal probability $1/m$, and each anchor position has probability $1/m$. A cell with an anchor or dummy endpoint has selection probability $1/m^2$, since that representative is independent of the other axis. The same holds for either root-to-root cell. Protect all such selected cells. The union bound gives a selection with no edited protected cell. We make no such probability assertion for general variable-variable cells or the shared leaf diagonal.

If B is K-free, every selected ordered role body must avoid P, by the unchanged five anchor incidences completing it to K. The root-plus diagonal is one and root-minus diagonal zero. Assuming these values at a parent, the two vertical bodies are

$$\begin{pmatrix} 1 & b_+ \\ 1 & 1 \end{pmatrix}, \qquad \begin{pmatrix} 1 & 0 \\ 1 & b_- \end{pmatrix}.$$

P-freeness forces $b_+ = 1, b_- = 0$. The horizontal bodies of the selected child parity are then

$$\begin{pmatrix} 1 & c_+ \\ 1 & 1 \end{pmatrix}, \qquad \begin{pmatrix} c_- & 0 \\ 1 & 1 \end{pmatrix},$$

forcing $c_+ = 1, c_- = 0$. All four bodies are ordered by the recursive tree order and have their dummy entries protected. Induction reaches the common leaf diagonal, a contradiction. Therefore $D_K(A_h) \geq m^2$. Edits to anchors, dummies and either binary label were never restricted.

7 Final quantifiers and blowups

Set $\epsilon_h = (8h+2)^{-2}$. For every fixed $C > 0$,

$$\frac{N_K(A_h)}{n_h^6 \epsilon_h^C} = 7(8h+2)^{2C-6} 4^{-h} \longrightarrow 0.$$

Given $c, C, \epsilon_0 > 0$, choose h with $\epsilon_h < \epsilon_0$ and this ratio $< c$. Replace each physical row and column by q consecutive identical twins. The pattern's distinct rows and columns prevent a copy from reusing a projected row or column, so its count multiplies exactly by q^6. The repair sampler works with blocks q times larger: protected-cell probabilities are $(qm)^{-2}$, and the lower bound is q^2m^2. Both normalized bounds are unchanged. Arbitrarily large q yields every prescribed lower threshold on host size at the **same** ϵ_h.

8 Verification and formal theorem

The original proof snapshot is retained byte for byte at the SHA-256 recorded in the separate-agent adversarial review. Sections 1–7 above reproduce its mathematical argument. The review found no gap and suggested two clarifications: the class-incidence paths meet at the leaf class, and the middle column in the W cases cannot be the dummy column because its middle-row entry is zero. These observations are already implied by the definitions. Separate-agent review is not human mathematical peer review.

The discovery census and literal checker agree through $h = 7$, including hosts of order 7,424. The latter checks expanded $h = 1$, three-seed repair guards through $h = 4$, all Boolean propagation states, and negative controls omitting each essential added support. The reviewer's independent implementation rebuilds the host from the prose, with complete weighted censuses through $h = 6$ and physical censuses through $h = 3$. Its optional SAT check establishes exact repair distances only at $h = 1$ and $h = 2$. Finite checks do not prove the universal statement or exact repair equality at all depths.

The principal Lean export is `OrderedRemoval.Crown.no_polynomial_removal_bound` in Complete.lean. It proves the qualitative theorem of Section 1 with the same literal 3×3 crown and n^6 normalization: every $c, C, \epsilon_0 > 0$ admits a fixed $0 < \epsilon < \epsilon_0$ such that every host-size threshold N admits an $n \geq \max(1, N)$ and a square binary host with at least ϵn^2 edits required in every crown-free binary repair and fewer than $c\epsilon^C n^6$ ordered copies. There are no host-existence, localization, sampling or anchor-preservation hypotheses.

For this certificate each anchor block on each axis is doubled to size $2m$, represented by two consecutive identical subgroups of size m. All body and dummy blocks, entry rules and relative block orders are retained. The formal host has $n = (14h + 2)m$. Localization bounds the number of eligible middle-zero cells by $2m$; encoding the other four indices gives at most $2mn^4$ copies. The protected sampler still gives a repair lower bound m^2. Consequently $\epsilon = (14h + 2)^{-2}$ and the density is at most $2\epsilon 2^{-h}$. Exponential decay and uniform blowups prove the full quantifier order. This certifies the qualitative 3×3 result, not the exact $7m^4$ count on the more economical written host.

The source-bound audit built and checked 92 local modules, 573 named theorems and 2,163 declarations, including inherited infrastructure. The Crown development adds 19 modules, 88 named theorems and 257 declarations. Only `propext`, `Quot.sound`, and `Classical.choice` occur as transitive axioms. Each local module was replayed into its imported Lean environment; the audit does not freshly replay all Mathlib or use a different proof assistant. Lean is pinned to 4.34.1 and Mathlib to `d13f23b723b8a846827a245b89c10fc7d3f11612`.

The executed-model check compares the actual Lean host and axis orders with a separate Python model through $h = 3$, covering 139,332 entries, and physical copy counts through $h = 2$. The formal semantic note is a same-agent second pass. The separate adversarial reviewer covered the written proof, not this Lean port. Exact repair equality, global minimality, and worldwide priority remain outside the certificate.

See formal/README.md and RUNLOG.md for reproducible commands, historical checkpoints and verification boundaries.

Acknowledgments

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Attribution

The signed tree, threshold entries, mode roles and protected-path argument in Sections 2 and 6 build on OpenAI's #190 manuscript and companion formalization [4]. The row supports adapt our 3×4 construction; the new changes are one anchor on each axis, suffix column anchors and vertical-dummy ones. Wang's manuscript [7] also gives the exact count of our preceding 4×4 refinement, with a different mode order. Further provenance and comparisons with the ProveIt and Erebos Labs constructions appear in BASELINE.md and ATTRIBUTION.md.

References

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